

AI Resume Rebuilder

Product Requirements Document (PRD) — V1.0

Platform: Web **Target Users:** Students, freshers, internship/job seekers

Product Vision: Upload an existing resume → extract and understand the information → optimize and restructure it → place it into a professional, ATS-friendly template → generate a polished PDF.

1. Product Vision

- **Core promise:** “Upload your resume. We rebuild it professionally.”
- The product is a student-first AI Resume Rebuilder rather than a generic resume builder. Students provide an existing PDF/DOCX resume and the system handles content extraction, organization, professional rewriting, template formatting, and PDF generation.

2. Problem Statement

- Students often have the required information but struggle with resume structure, wording, project descriptions, section ordering, formatting, and ATS-friendly presentation.
- Traditional resume builders make students manually solve these problems. This product reverses that workflow: upload an existing resume and let the system rebuild it.

3. Target Audience

- **Primary:** B.Tech/BE, BCA, MCA, BBA, MBA, diploma and other college students and freshers.
- **Secondary:** Internship applicants, recent graduates and entry-level developers.

4. Core User Journey

- Landing Page → Upload Resume → Processing → Information Extraction → Review Extracted Data → Select Target Role → AI Optimization → Choose Template → Live Preview → Edit → Resume Score → Generate PDF → Download.

5. V1 Features

- **Landing page:** clear value proposition, upload CTA and example resume.
- **Authentication:** Google Sign-In and email/password using Firebase Authentication.
- **Resume upload:** PDF and DOCX, initially up to 5 MB and 3 pages.
- **Resume processing:** extract text and identify structured resume sections.
- **Editable extracted data:** users can correct AI/parser mistakes before final generation.
- **Target role:** select roles such as Android Developer, Java Developer, Software Engineer, Full Stack Developer, Data Analyst, etc.
- **AI optimization:** improve wording, summaries and project bullets without inventing facts.
- **Templates:** three initial templates — ATS Classic, Developer, Fresher.
- **Live preview:** desktop split editor/preview and responsive mobile layout.
- **Resume score:** completeness, structure, skills, projects, contact information, ATS compatibility and role relevance.
- **PDF download:** deterministic HTML/CSS template rendered to PDF.

6. Information Extraction

- **Personal:** name, email, phone, location, LinkedIn, GitHub, portfolio.
- **Summary:** existing professional summary/objective.

- **Education:** degree, branch, institution, dates, CGPA/percentage.
- **Skills:** programming languages, frameworks, libraries, databases, tools and technologies.
- **Projects:** name, description, technologies, features and links.
- **Experience:** company, role, dates and responsibilities.
- **Certifications:** name, issuer, date and credential URL.
- **Achievements:** achievement, organization and date.
- V1 should avoid forcing irrelevant sections; only include sections supported by the source data.

7. Internal Resume Schema

- AI should produce structured data, not HTML. The canonical model contains personalInfo, summary, skills, education, projects, experience, certifications and achievements.

Example JSON:

```
{
  "personalInfo": {
    "name": "...",
    "email": "...",
    "summary": "...",
    "skills": {
      "languages": [],
      "frameworks": [],
      "databases": [],
      "tools": []
    },
    "education": [],
    "projects": [],
    "experience": [],
    "certifications": [],
    "achievements": []
  }
}
```

8. AI Optimization Rules

- AI may rewrite content professionally while preserving factual meaning.
- **Strict rule:** never invent companies, technologies, responsibilities, achievements, metrics, dates or project features.
- Existing good summaries should be improved rather than unnecessarily replaced. If no summary exists, generate a concise 2–4 line summary from verified resume information.
- Project descriptions should be converted into strong, concise resume bullets while preserving the original facts.

9. Template System

- **ATS Classic:** minimal, traditional structure.
- **Developer:** technical-skills and projects focused.
- **Fresher:** education, skills, projects, internships, certifications and achievements focused.
- **Architecture rule:** AI controls content; application controls structure; CSS controls design; PDF engine controls output.
- AI must never generate the complete resume HTML.

10. Firebase Architecture

- **Firebase Authentication:** login/signup.
- **Cloud Firestore:** users, resume metadata and structured resume data.
- **Cloud Storage:** original uploaded resumes and generated PDFs.
- **Firebase App Check:** protect Firebase resources.
- **Firebase Analytics:** product usage and funnel analytics.
- **Optional:** Firebase Hosting for frontend deployment and Cloud Functions for future event-driven tasks.

11. Technology Stack

- **Frontend:** Next.js, TypeScript, Tailwind CSS, shadcn/ui, React Hook Form, Zod, Firebase SDK.
- **Backend:** Java 21, Spring Boot, Spring Web, Spring Validation, Firebase Admin SDK, Maven, Jackson, Lombok.
- **Document processing:** Apache Tika and appropriate PDF/DOCX parsers.
- **AI:** an external AI API accessed through Spring Boot.

- **PDF:** server-side HTML/CSS-to-PDF generation.

12. Backend Architecture

- Next.js → Firebase Authentication → Firebase ID Token → Spring Boot → parser/AI/template/PDF services → Firestore/Storage.
- Recommended packages: auth, resume (controller/service/repository/model/dto), parser, ai, template, pdf, firebase and common/config.

13. Firestore Data Model

- users/{userId}: name, email, photoURL, createdAt.
- resumes/{resumeId}: userId, originalFile, targetRole, templateId, status, score, createdAt, updatedAt.
- Keep V1 simple; expand into subcollections only when document size or product requirements justify it.

14. API Design

- GET /api/v1/users/me
- POST /api/v1/resumes
- GET /api/v1/resumes
- GET /api/v1/resumes/{id}
- PUT /api/v1/resumes/{id}
- DELETE /api/v1/resumes/{id}
- POST /api/v1/resumes/{id}/upload
- POST /api/v1/resumes/{id}/process
- POST /api/v1/resumes/{id}/optimize
- GET /api/v1/templates
- POST /api/v1/resumes/{id}/generate-pdf
- GET /api/v1/resumes/{id}/download

15. Processing Status

- UPLOADING → PROCESSING → EXTRACTING → OPTIMIZING → READY.
- On failure, status becomes FAILED with a user-readable error message.

16. Security & Privacy

- Use Firebase Authentication and App Check, authenticated Firestore/Storage rules, Spring Boot Firebase token verification, HTTPS, file type/size validation, ownership checks and AI rate limiting.
- Never trust a client-provided user ID. Derive identity from the verified Firebase token.
- Resume files must not be publicly accessible.
- Provide clear retention/deletion information and controls to delete resumes and eventually the full account/data.

17. Analytics

- Track: landing_page_view, resume_upload_started, resume_upload_completed, resume_processing_completed, resume_processing_failed, target_role_selected, template_selected, resume_edited, resume_score_viewed, pdf_generated and pdf_downloaded.
- Primary funnel metric: uploaded resume → successfully processed → edited → PDF generated.

18. MVP Scope

- **Frontend:** landing page, authentication, upload, processing screen, extracted-data review, target-role selection, editor, three templates, live preview, score, PDF download and dashboard.
- **Backend:** Firebase token verification, resume CRUD, file processing, extraction, AI optimization, template rendering and PDF generation.
- **Firebase:** Authentication, Firestore, Storage and App Check.
- Do not add payments, job matching, LinkedIn import or advanced ATS matching until the core pipeline is stable.

19. V2 Roadmap

- More templates, one-click template switching, advanced ATS analysis, job-description matching, keyword optimization, LinkedIn/GitHub imports and stronger AI-assisted improvements.

20. V3 Vision

- Expand into a broader student career platform: resume builder, job matching, portfolio builder, cover-letter generation and interview preparation.

21. Monetization

- Potential freemium model: free basic resume/template and paid access to unlimited resumes, premium templates, AI optimization, job-specific optimization and advanced ATS analysis.
- Suggested future student-friendly pricing: approximately \$99–\$199/month or a low-cost one-time purchase. Do not implement payments in V1.

22. Key Product Principle

- **AI understands the student's resume; our application decides what the final resume looks like.**
- This keeps the product consistent, safer against hallucinations, easier to test, and capable of producing deterministic professional templates.

Recommended Architecture

Layer	Technology	Responsibility
Web	Next.js + TypeScript	UI, editor, preview, routing
Auth	Firebase Auth	Google/email authentication
API	Spring Boot + Java 21	Business logic and secure API
Database	Cloud Firestore	Resume/user metadata and structured data
Files	Firebase Storage	Original resumes and generated PDFs
AI	AI API via Spring Boot	Extraction, normalization and optimization
Templates	HTML/CSS	Deterministic professional resume designs
PDF	Server-side renderer	Final downloadable PDF
Analytics	Firebase Analytics	Funnel and product usage